# KRISHNASAI ADDALA

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## SUMMARY

AI Engineer building production-first GenAI infrastructure — reliable, scalable agentic systems for enterprise deployment, spanning retrieval, orchestration, evaluation, and monitoring. Shipped LangGraph-based GenAI tools to financial institution clients on GCP, owned LLMOps and multi-environment deployment pipelines, and delivered predictive models to energy-sector stakeholders. Independently built an LLM infrastructure and interpretability evaluation platform re-architected for production reliability, plus a large-scale multi-agent governance simulation system.

## EXPERIENCE

**Capgemini** — *New York City, NY*
*Associate Consultant — GenAI Developer* — Jun 2025 – Nov 2025

- Built and shipped AI-powered compliance and customer-satisfaction tools for financial institution clients using LangGraph and OpenAI API, delivered to production and integrated into client workflows.
- Engineered scalable ETL and streaming pipelines on GCP (Airflow, BigQuery, Pub/Sub, Cloud Functions) for ingesting live data into centralized ML environments powering downstream decision-support systems.
- Designed multi-stage NLP processing systems spanning retrieval, validation, orchestration, and automated monitoring, improving analytical consistency and system reliability while cutting manual review overhead.
- Owned LLMOps practices — model versioning, multi-environment deployment (dev to staging to production), and Airflow-orchestrated release pipelines — while collaborating across engineering and client-facing teams to operationalize ML workflows at scale.

**Project Geminae — MSBA Capstone** — *Pittsburgh, PA*
*Machine Learning Engineer* — Jan 2025 – Apr 2025

- Delivered predictive well-selection models for an energy client, enabling prioritization of oil-well candidates across 30+ operational variables.
- Built a rigorous feature engineering pipeline — correlation analysis, leakage prevention, benchmarking — that measurably improved held-out predictive performance over baseline.
- Generated feature importance outputs that surfaced key productivity drivers in engineer-readable format, directly supporting stakeholder decision-making.

**MIDAS Lab, IIIT Delhi** — *New Delhi, India*
*Undergraduate Research Engineer* — Jun 2023 – May 2024

- Shipped a knowledge-graph-augmented LLM system for physics question answering that achieved ~200% improvement over the baseline and contributed to three published papers.
- Built structured evaluation pipelines for benchmarking system performance across retrieval, reasoning, and answer-quality dimensions.
- Coordinated three cross-functional ML sub-teams across retrieval systems, model development, dataset curation, and benchmarking — maintained delivery across parallel workstreams.

## INDEPENDENT ENGINEERING

**AI Governance Policy Simulator** — Dec 2025 – Jun 2026
*Python · NumPy · SciPy · NetworkX · Agent-Based Modeling*

- Built a production-grade multi-agent simulation platform at scale — 1000+ agents across 30 experimental runs — modeling AI policy enforcement and discourse dynamics across synthetic social networks.
- Implemented a counterfactual experiment engine identifying a geometry-metric-dependent sign reversal that flips predicted policy outcomes, a finding with direct implications for real AI governance frameworks.
- Designed modular enforcement, shock-propagation, and hub-targeting subsystems enabling parameterized scenario testing; paper in preparation for arXiv submission.

**LLM Infrastructure & Interpretability Platform** — Feb 2026 – Present
*Python · FastAPI · SSE · PyTorch · CUDA · HuggingFace · MLflow · Docker*

- Built an end-to-end behavioral auditing and mechanistic interpretability platform for open-weight LLMs (Qwen2.5, GPT-2), purpose-built to surface trust-critical failure modes in agentic and tool-calling deployments.

- Re-architected the evaluation server from a two-process file-IPC design to a single-process FastAPI + SSE-streaming architecture, eliminating race conditions and cutting evaluation latency by ~60% — production-reliability engineering applied to LLM infrastructure.
- Behavioral layer: 9 AI safety and behavioral probe suites covering jailbreak resistance, toxicity, hallucination, and power-seeking, run through the production evaluation server.
- Interpretability layer: built a purpose-built Natural Language Autoencoder for observing small models (≤7B parameters), with per-token residual stream extraction, attention metric capture, and cross-model geometric similarity metrics — exposing the internal state driving behavioral failures in tool selection and multi-step planning.
- Key finding: geometry/function decoupling in transformer residual streams is extreme; text-as-description conditioning outperforms heuristic labels for latent reconstruction, with direct implications for diagnosing hallucination and miscalibration in production agents.

## TECHNICAL SKILLS

**Languages:** Python, TypeScript, JavaScript, SQL, R, Java, Bash

**Agentic Development:** LangChain, LangGraph, Google ADK, OpenAI API, HuggingFace Transformers, RAG, Vector Stores, Agentic Workflows, Tool Use

**ML:** PyTorch, TensorFlow, Scikit-learn, XGBoost, LightGBM, CUDA, JAX, SHAP, LIME, WandB

**Cloud:** GCP (BigQuery, Pub/Sub, Cloud Functions, Vertex AI, Airflow), AWS (EC2, S3), Docker, Kubernetes, MLflow

**Backend & APIs:** FastAPI, Flask, REST APIs, Streamlit, ReactJS, SSE Streaming

**Data:** Pandas, NumPy, Spark, Databricks, PostgreSQL, MongoDB, Neo4j, Power BI, Tableau

## EDUCATION

**Carnegie Mellon University — Tepper School of Business** May 2025
*M.S. Business Analytics (MSBA)* | *Pittsburgh, PA*

**IIIT Delhi** May 2024
*B.Tech Computer Science, Specialization: Computational Social Science* | *New Delhi, India*

## PUBLICATIONS

- Knowledge Graphs are all you need: Leveraging KGs in Physics Question Answering — arXiv, Dec 2024 (arxiv.org/abs/2412.05453)
- Steps are all you need: Rethinking STEM Education with Prompt Engineering — arXiv, Dec 2024 (doi.org/10.48550/arXiv.2412.05023)
- Revolutionizing High School Physics Education: A Novel Dataset — Big Data and AI Conference, Dec 2023 (Springer)